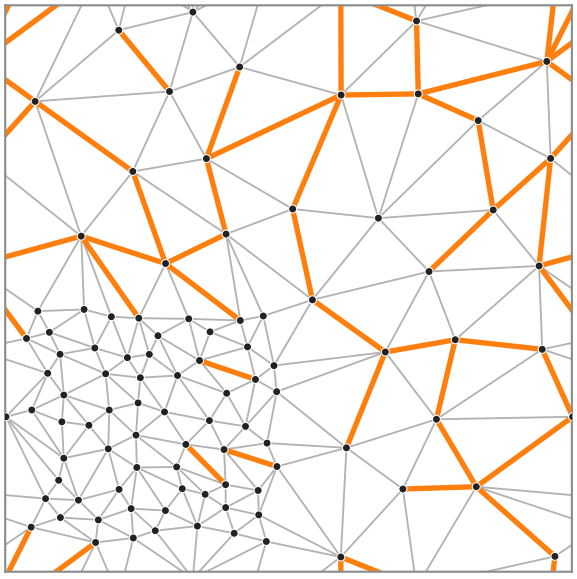
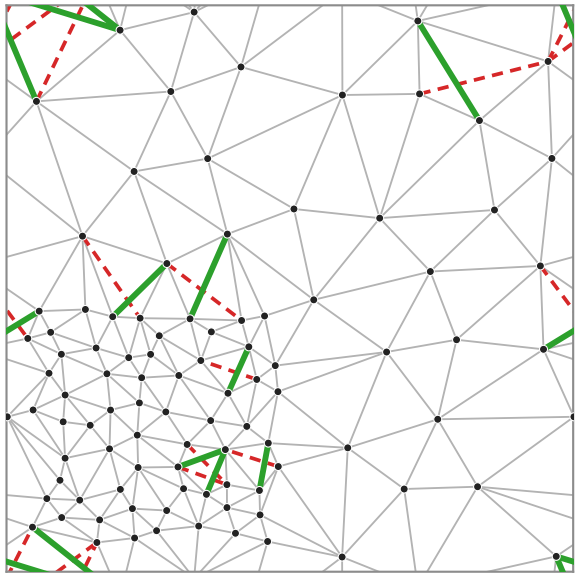
7 Screen GPU



One pass over the old triangles at the new points: the in-circle test of each edge in double, with a rounding bound. Failing, unsure, and (in so small a box) long edges are listed: 42 of 288.

`tri_repair.hpp: screen`
`3D: tet_repair.hpp: move_screened`

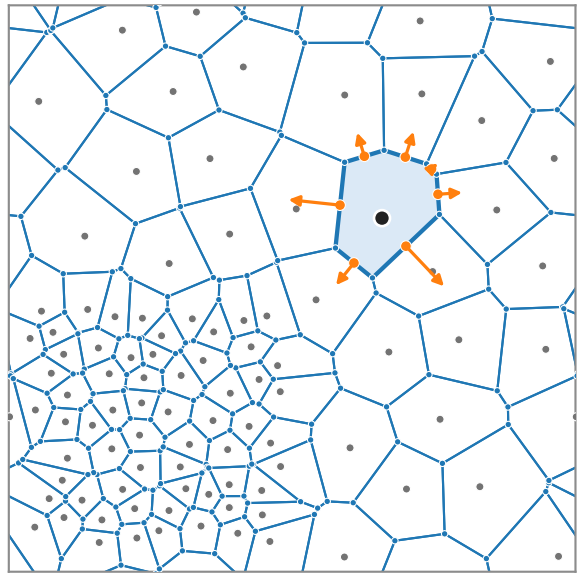
8 Flip GPU



The exact test decides each listed edge. Flips (dashed: gone, green: new) run in rounds; votes settle conflicts, so thread order is moot. 13 flips in 6 rounds; volumes equal a fresh build's.

`tri_repair.hpp: repair`
`3D: tet_repair.hpp: flip23, flip32`

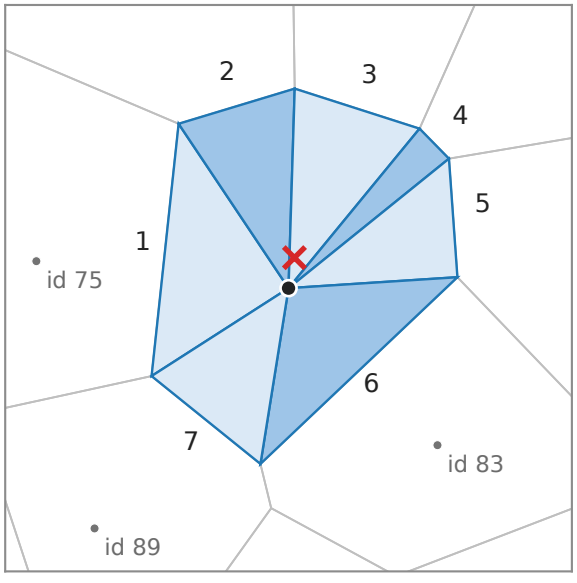
9 Geometry: the dual GPU



Circumcentres (blue dots) are the cell corners: the Voronoi mesh is the dual of the triangles. One thread per face writes its area vector (arrows) and centroid (orange).

`tri_mesh.hpp: faces`
`3D: tet_mesh.hpp: faces`

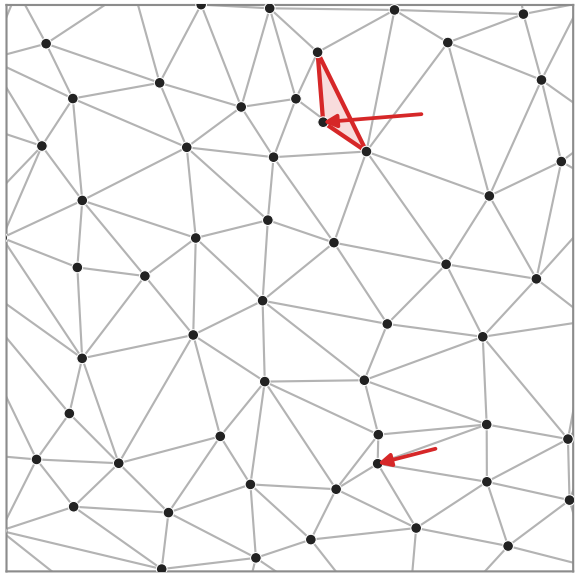
10 update_cells() GPU



Per cell, a list of its faces sorted by the other particle's id (1 … 7). Volume and centroid (×) are summed over the pyramids in that order: the same bits on every run. Σ volume = L^2 exactly here.

`tet_cells.hpp: cells`
`(the same code in 2D and 3D)`

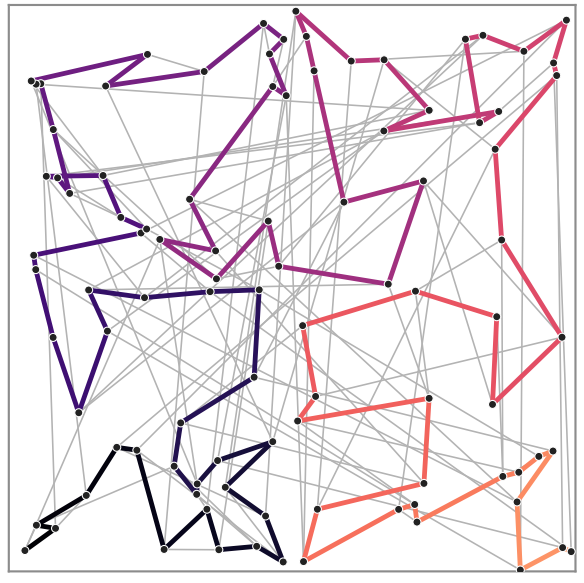
11 Fallbacks host GPU



Two particles jump (arrows); flips leave 1 inverted triangle (red). Few stuck: the host redoes a patch (11 triangles), the GPU flips again. Many: a rebuild (1–5). Two on one grid point: refused.

`tri_finish.hpp: finish (host)`
`3D: tet_flips.hpp: finish`

12 Re-sort (3D) GPU



Over many steps ids drift apart in space (grey: id order). Renumbering along a Hilbert curve (colour) restores locality: the mean id gap over faces is 13 at the start, 24 now, 13 after.

`mesh_order.hpp: cell_order`
`voronoi_mesh.hpp: Mesh::resort`